Macroeconomics #1

Some Mathematical Preliminaries

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Today's lecture

1. Static Optimization
2. Dynamic Optimization
3. Optimal Control (optional)
4. Differential Equations and Phase Diagrams
5. Appendix

Review of Static Constrained Optimization

From unconstrained to constrained problems

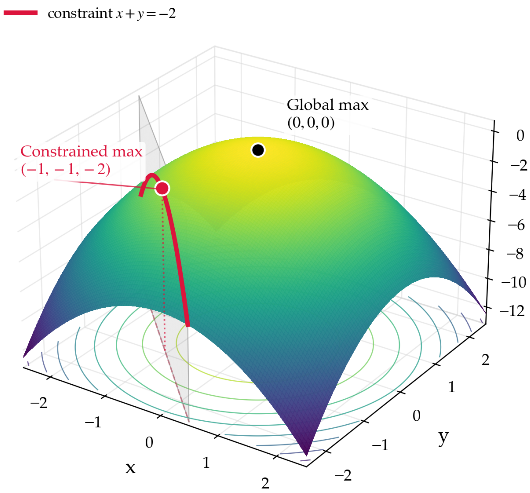
In most economic problems the planner is *not* free to choose x anywhere: resources, budgets and technologies impose constraints. Compare

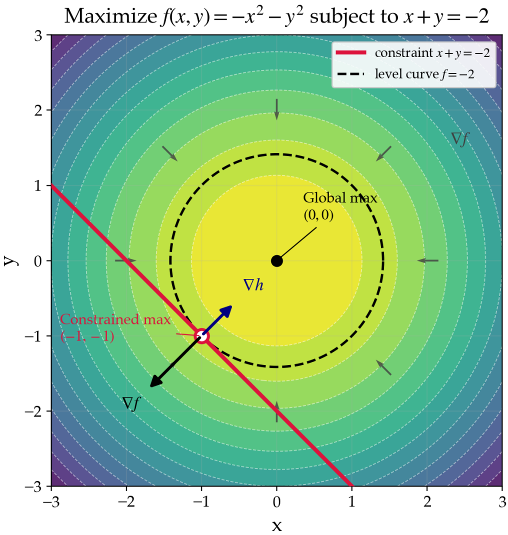
$$\max_x f(x) \qquad \text{versus} \qquad \max_x f(x) \;\; \text{s.t. } h_j(x) = c_j, \; j = 1, \dots, k.$$

- In the unconstrained problem we can set $\nabla f(x^*) = 0$, because we may move in every direction.
- With constraints, the optimum need *not* be a critical point of f: we may only move *along* the constraint set.
- The **Lagrange multiplier method** turns the constrained problem into an unconstrained one in more variables.

Standard case: $x \in \mathbb{R}^n$, k equality constraints, with $n > k$.

- Unconstrained: $\max -x^2 - y^2$
- Constrained: $\max -x^2 - y^2$ s.t. $x + y = -2$




Maximize $f(x, y) = -x^2 - y^2$ subject to $x + y = -2$
constraint $x + y = -2$
level curve $f = -2$
∇f
Global max
$(0, 0)$
∇h
Constrained max
$(-1, -1)$
∇f
y
x
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

At the optimum

Consider two variables and one constraint,

$$\max_{x,y} f(x,y) \quad \text{s.t.} \quad h(x,y) = c.$$

Along the constraint, $y = y(x)$, so the objective is $f(x, y(x))$. At an interior optimum,

$$\frac{d}{dx} f(x, y(x)) = f_x + f_y \, y'(x) = 0.$$

Differentiating the constraint, $h_x + h_y \, y'(x) = 0$, so $y'(x) = -h_x / h_y$. Substituting,

$$\frac{f_x}{h_x} = \frac{f_y}{h_y} = \lambda.$$

In other words

$$f_x - \lambda h_x = 0,$$
$$f_y - \lambda h_y = 0$$

Lagrangian

Reformulate the original problem from constrained to unconstrained case by rewriting the objective function to a Lagrangian function

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda[c - h(x, y)]$$

FOC

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= f_x - \lambda h_x = 0, \\ \frac{\partial \mathcal{L}}{\partial y} &= f_y - \lambda h_y = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= c - h(x, y) = 0 \end{aligned}$$

For the Lagrangian to work, the constraint gradient must not vanish: $(h_x, h_y) \neq (0, 0)$ at the optimum.

The Lagrangian and the FOC

Problem. Objective $f(x)$, $x \in \mathbb{R}^n$; constraints $h_j(x) = c_j$, $j = 1, \ldots, k$, with $n > k$.

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad h_j(x) = c_j,\ j = 1, \ldots, k.$$

Lagrangian. Introduce one multiplier per constraint:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^{k} \lambda_j \left(c_j - h_j(x)\right).$$

First-order necessary conditions

$$\frac{\partial \mathcal{L}}{\partial x_i} = 0 \ (i = 1, \ldots, n), \qquad \frac{\partial \mathcal{L}}{\partial \lambda_j} = 0 \iff h_j(x) = c_j.$$

This converts the constrained problem into an *unconstrained* one in $(n + k)$ variables. (Regularity: the $\nabla h_j(x^*)$ must be linearly independent.)

Example 1: two-period consumption

A household lives for two periods and maximizes

$$\max_{c_1, c_2} U(c_1, c_2) = c_1 c_2 \quad \text{s.t.} \quad c_1 + s = B, \qquad c_2 = (1+r)s,$$

with $B > 0$ and $r > 0$.

- Eliminate s: $s = B - c_1$, so the budget becomes

$$c_1 + \frac{c_2}{1+r} = B.$$

- We solve the same problem twice: by substitution and by the Lagrangian.

Example 1: solution by substitution

Use the budget constraint to eliminate c_2:

$$c_2 = (1+r)(B - c_1).$$

Substitute into the objective:

$$U(c_1) = c_1(1+r)(B - c_1) = (1+r)(Bc_1 - c_1^2).$$

FOC:

$$U'(c_1) = (1+r)(B - 2c_1) = 0 \quad \Longrightarrow \quad c_1^* = \frac{B}{2}.$$

SOC:

$$U''(c_1) = -2(1+r) < 0,$$

so this is a maximum. Then

$$c_2^* = (1+r)\left(B - \frac{B}{2}\right) = \frac{(1+r)B}{2}.$$

Example 1: solution by Lagrangian

The Lagrangian is

$$\mathcal{L}(c_1, c_2, \lambda) = c_1 c_2 + \lambda \left(B - c_1 - \frac{c_2}{1+r} \right).$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial c_1} = c_2 - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial c_2} = c_1 - \frac{\lambda}{1+r} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = B - c_1 - \frac{c_2}{1+r} = 0.$$

From the first two equations, $c_2 = \lambda$ and $c_1 = \lambda/(1+r)$, so $c_2 = (1+r)c_1$. Substituting into the budget constraint,

$$c_1 + \frac{(1+r)c_1}{1+r} = 2c_1 = B \implies c_1^* = \frac{B}{2}, \qquad c_2^* = \frac{(1+r)B}{2},$$

$$\lambda^* = \frac{(1+r)B}{2}.$$

Same solution as substitution. The multiplier equals the marginal utility of wealth; the next slide verifies the SOC with the bordered Hessian.

SOC: the bordered Hessian (2 variables, 1 constraint)

Curvature must be checked *along the constraint*. Border $\nabla^2\mathcal{L}$ with the constraint gradient:

$$\overline{\mathbf{H}} = \begin{pmatrix} 0 & h_x & h_y \\ h_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ h_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{pmatrix}_{(x^*,y^*,\lambda^*)}.$$

$$|\overline{\mathbf{H}}| > 0 \Rightarrow \text{local maximum}, \qquad |\overline{\mathbf{H}}| < 0 \Rightarrow \text{local minimum}.$$

Example 1. With $h = c_1 + c_2/(1+r) - B$,

$$|\overline{\mathbf{H}}| = \begin{vmatrix} 0 & 1 & 1/(1+r) \\ 1 & 0 & 1 \\ 1/(1+r) & 1 & 0 \end{vmatrix} = \frac{2}{1+r} > 0,$$

so (c_1^*, c_2^*) is a maximum. Intuition: only tangent directions ($\nabla h \cdot d = 0$) are admissible, and $\overline{\mathbf{H}}$ restricts the second-order test to exactly those directions.

SOC: general case

For n variables and k constraints ($n > k$), border $\nabla^2 \mathcal{L}$ with the k constraint gradients:

$$\overline{\mathbf{H}} = \left(\begin{array}{c|c} \mathbf{0}_{k\times k} & \left[\dfrac{\partial h_j}{\partial x_i} \right] \\ \hline \left[\dfrac{\partial h_j}{\partial x_i} \right]^T & \left[\dfrac{\partial^2 \mathcal{L}}{\partial x_i \, \partial x_\ell} \right] \end{array} \right).$$

Check the last $n - k$ *bordered leading principal minors* $|\overline{H}_{k+1}|, \ldots, |\overline{H}_n|$:

- Minimum: all share the sign of $(-1)^k$.
- Maximum: signs alternate, with $(-1)^n |\overline{H}_n| > 0$.
- Any other pattern $\Rightarrow$ neither a max nor a min.

Special case $k = 1$: min if all $|\overline{H}_i| < 0$; max if they alternate.

The meaning of the multiplier

The multiplier measures how the *optimal value* changes when the constraint is relaxed:

$$\max_x f(x) \;\text{ s.t. }\; h(x) = c \qquad \Longrightarrow \qquad \frac{df^*(c)}{dc} = \lambda.$$

For a small change Δc,

$$f^*(c + \Delta c) - f^*(c) \approx \lambda(c)\,\Delta c.$$

Thus λ is the **shadow price** of the constraint. In Example 1, $\lambda^* = (1+r)B/2$ is the marginal utility of wealth: an extra unit of B raises utility by λ^*.

Example 2: representative agent

A representative agent consumes x and chooses hours worked ℓ:

$$\max_{x,\ell} U(x,\ell) = \alpha \ln x + \beta \ln(1-\ell) \quad \text{s.t.} \quad px = w\ell + b,$$

with $0 \leq \ell \leq 1$ and $x, p, w, \alpha, \beta > 0$; b is non-labour income.
Find the optimal choices of x, ℓ and verify they are indeed optimal by constructing a bordered Hessian matrix.

Dynamic Optimization

The Cake Eating Problem

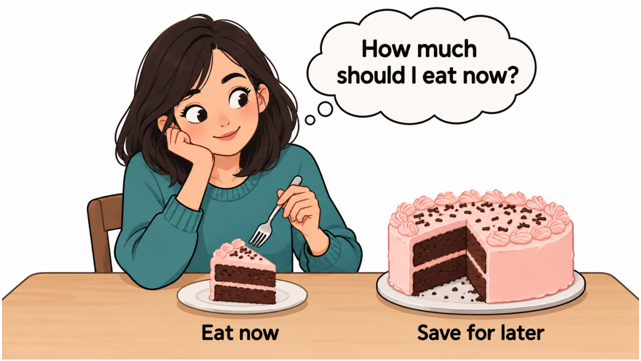

How much should I eat now?
Eat now
Save for later

Figure: AI-generated

Infinite horizon planner: cake-eating problem

Initial cake stock $k_0 > 0$. We have the following transition equation:

$$k_{t+1} = k_t - c_t, \qquad c_t \geq 0,\; k_{t+1} \geq 0.$$

Planner chooses $\{c_t\}_{t\geq 0}$ to solve the lifetime utility:

$$\max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

- State variable: k_t (the remaining cake).
- Control variable: c_t (current consumption).
- Instantaneous utility: $u(c_t)$

$(*)$ To be rigorous, the max operator should be replaced by sup as we cannot guarantee that the optimum exists. For simplicity, we assume regularity conditions under which a maximum exists and is unique (e.g. strictly concave u).

Lagrangian approach

The Lagrangian:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (k_t - c_t - k_{t+1})]$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t [u'(c_t) - \lambda_t] = 0 \iff \lambda_t = u'(c_t),$$
$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = -\beta^t \lambda_t + \beta^{t+1} \lambda_{t+1} = 0 \iff \lambda_t = \beta \lambda_{t+1}$$

Substitute $\lambda = u'(c_t)$ to obtain the Euler equation

$$u'(c_t) = \beta u'(c_{t+1})$$

The solution is a sequence of consumption path $\{c_t\}_{t=0}^{\infty}$.

Recursive Structure

By define the value function given an initial cake size k

$$V(k) = \max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t. } k_{t+1} = k_t - c_t,\ k_0 = k \tag{1}$$

We obtain maximum lifetime utility starting with cake. Assuming that all future optimal values have been found, the objective function can be written as

$$\sum_{t=0}^{\infty} \beta^t u(c_t) = u(c_0) + \beta \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \tag{2}$$

or (suppressing the time notation)

$$V(k) = u(c) + \beta V(k - c) \tag{3}$$

This exploits the principle of optimality: An optimal plan from today must remain optimal tomorrow and any future point onward.

Bellman formulation of cake-eating

The Bellman equation:

$$V(k) = \max_{0 \leq c \leq k} \{u(c) + \beta V(k-c)\}.$$

Equivalent choice in next-state form:

$$V(k) = \max_{0 \leq k' \leq k} \{u(k-k') + \beta V(k')\}.$$

This is the canonical one-state dynamic programming model.

- state variable: k
- choice variable: k'

FOC and envelope for interior solution

FOC:

$$u'(c) = \beta V'(k').$$

Envelope condition:

$$V'(k) = u'(k - k') = u'(c)$$

Shift envelope one period:

$$V'(k') = u'(c').$$

So Euler equation becomes

$$u'(c) = \beta u'(c').$$

Euler equation

The same Euler equation is obtained

$$u'(c_t) = \beta u'(c_{t+1})$$

Intuition:

- LHS: marginal utility **loss** from reducing c_t by one unit
- RHS: discounted marginal utility **gain** from adding that unit to tomorrow.

The agent obtain optimal consumption at the point where LHS=RHS. If u is concave, planner smooths consumption over time.

Closed-form solution for log utility

Let $u(c) = \log c$. Euler:

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \Rightarrow c_{t+1} = \beta c_t.$$

So consumption is geometric:

$$c_t = \beta^t c_0.$$

Resource feasibility requires

$$\sum_{t=0}^{\infty} c_t = k_0$$

This implies

$$c_0(1 + \beta + \beta^2 + \cdots) = k_0 \tag{4}$$

Applying geometric sequence

$$c_0 = (1 - \beta)k_0 \tag{5}$$

Policy and value functions for log utility

The policy function is a function taking the state variable as input to output the choice/control variable

$$c_t = h(k_t) \tag{6}$$

Optimal policy:

$$c_t^* = (1-\beta)k_t \qquad k_{t+1}^* = \beta k_t.$$

We can also solve for the Value function $V(k)$ by Guess-and-Verify. Assume that the value function takes the form :

$$V(k) = A + B \log k.$$

This guess is reasonable because the period utility is logarithmic and the constraint is linear in k, so the value function should inherit the log form. One can solve for A, B as ▸ See derivation

$$B = \frac{1}{1-\beta}, \qquad A = \frac{\log(1-\beta)}{1-\beta} + \frac{\beta \log \beta}{(1-\beta)^2}.$$

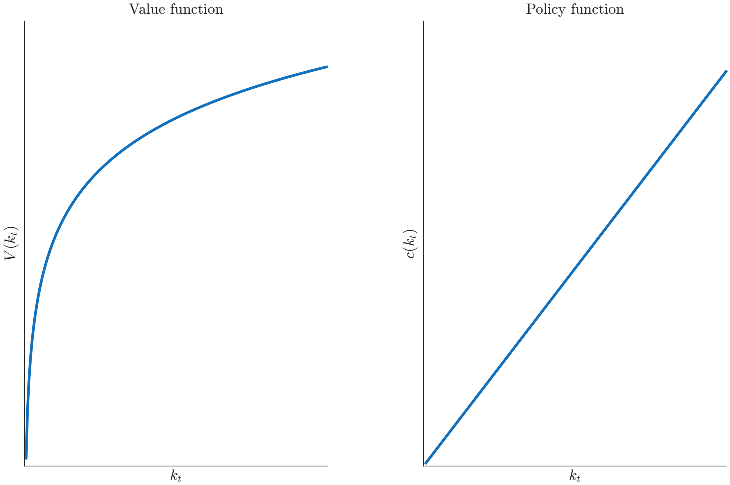


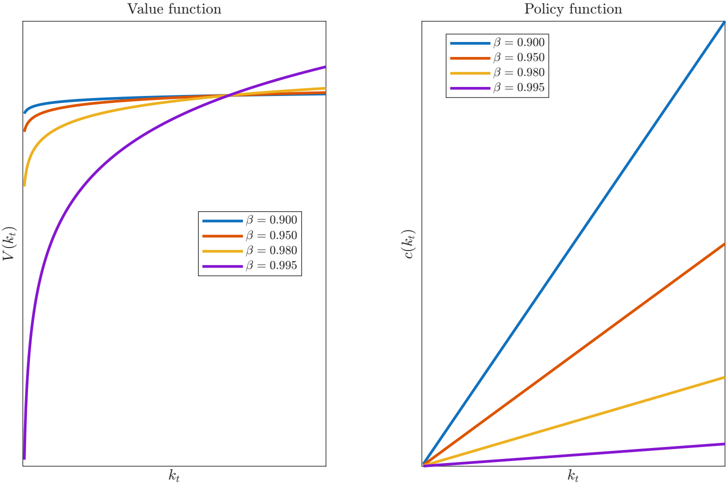
Figure: Value function and Policy function as functions of k_t

Comparative statics:

$$c_0 = (1-\beta)k_0, \qquad k_{t+1} = \beta k_t.$$

- Higher β (more patient): lower initial consumption, slower cake depletion.
- Lower β: front-loaded consumption and faster depletion.

However, the Euler equation is not sufficient to guarantee optimality. It does not rule out the paths that satisfy the Euler equation but leave valuable cake unconsumed forever. When it happens, we could increase consumption slightly in the future and improve utility, but then, the chosen consumption path will no longer optimal.
Therefore, we need a transversality condition to rule out those paths.


Value function
$V(k_t)$
k_t
$\beta = 0.900$
$\beta = 0.950$
$\beta = 0.980$
$\beta = 0.995$
Policy function
$c(k_t)$
k_t
$\beta = 0.900$
$\beta = 0.950$
$\beta = 0.980$
$\beta = 0.995$

Deriving the transversality condition

Start with a finite horizon T and impose $k_{T+1} \geq 0$ (multiplier $\beta^T \mu$):

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \Big[u(c_t) + \lambda_t (k_t - c_t - k_{t+1}) \Big] + \beta^T \mu \, k_{T+1}.$$

FOCs: $u'(c_t) = \lambda_t$ for all t, $\lambda_t = \beta \lambda_{t+1}$ for $t < T$, and for the last choice k_{T+1}:

$$-\beta^T \lambda_T + \beta^T \mu = 0, \qquad \mu \geq 0, \qquad \mu \, k_{T+1} = 0.$$

Hence $\mu = \lambda_T = u'(c_T)$ and complementary slackness reads

$$\beta^T u'(c_T) \, k_{T+1} = 0.$$

Since $u'(c_T) > 0$, it is optimal to leave **no cake** at the end: $k_{T+1} = 0$.

Deriving the transversality condition

Infinite horizon. The transversality condition is the limit of the terminal condition:

$$\lim_{T\to\infty} \beta^T u'(c_T)\, k_{T+1} = 0.$$

Why it is needed. Suppose a plan satisfies the Euler equation but $\lim_{T\to\infty} \beta^T u'(c_T) k_{T+1} > 0$. Then cake with a positive (discounted) value is never eaten. Eating a little more of it at some date raises utility without violating any constraint, so the plan cannot be optimal.

Interpretation:

- k_{T+1}: leftover cake
- $\beta^T u'(c_T)$: value of a unit of cake, in today's utility

The TVC ensures that **no** valuable resources are left unused at infinity.

Transversality condition in cake-eating

For this model:

$$\lim_{t\to\infty} \beta^t u'(c_t) k_{t+1} = 0.$$

With log utility and optimal policy:

$$\beta^t \frac{1}{c_t} k_{t+1} = \beta^t \frac{1}{(1-\beta)\beta^t k_0} (\beta^{t+1} k_0) = \frac{\beta^{t+1}}{1-\beta}.$$

And

$$\lim_{t\to\infty} \frac{\beta^{t+1}}{1-\beta} = 0$$

since $\beta < 1$, the TVC is satisfied.

It's better to see the role of TVC when there is another suboptimal plan

Assumption: $\beta = 0.9$, $k_0 = 1$, $u(c) = \ln c$.
The optimal plan (Euler equation + TVC) is:

$$\begin{aligned} c_t &= (1-\beta)k_t, \\ k_t &= \beta^t k_0 \end{aligned}$$

Now, compare the optimal plan with a half-cake plan:

	Optimal plan	Half-cake plan
Consumption	$c_t = 0.1(0.9)^t$	$c_t = 0.05(0.9)^t$
Cake left	$k_{t+1} = 0.9^{t+1}$	$k_{t+1} = 0.5 + 0.5(0.9)^{t+1}$
Limit cake	$k_\infty = 0$	$k_\infty = 0.5$
TVC limit	0	10

Both satisfy the Euler equation:

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}}.$$

The optimal plan: eat 10% of the cake each period

$$c_t^* = 0.1(0.9)^t, \qquad k_{t+1}^* = 0.9^{t+1}.$$

t	eat c_t	left k_{t+1}	$\ln c_t$	$\beta^t \ln c_t$	TVC term
0	0.1000	0.9000	−2.3026	−2.3026	9.0000
1	0.0900	0.8100	−2.4079	−2.1672	8.1000
2	0.0810	0.7290	−2.5133	−2.0358	7.2900
3	0.0729	0.6561	−2.6187	−1.9090	6.5610
4	0.0656	0.5905	−2.7240	−1.7872	5.9049
5	0.0590	0.5314	−2.8294	−1.6707	5.3144
6	0.0531	0.4783	−2.9347	−1.5596	4.7830
7	0.0478	0.4305	−3.0401	−1.4541	4.3047
8	0.0430	0.3874	−3.1455	−1.3540	3.8742
9	0.0387	0.3487	−3.2508	−1.2594	3.4868
10	0.0349	0.3138	−3.3562	−1.1702	3.1381
∞	0	0	—	—	0

The cake is exhausted asymptotically: $k_t \to 0$. The TVC term is

$$\beta^t u'(c_t) k_{t+1} = 10\beta^{t+1} \to 0.$$

The half-cake plan: eat half of every optimal slice

$$c_t = 0.05(0.9)^t, \qquad k_{t+1} = 0.5 + 0.5(0.9)^{t+1}.$$

t	eat c_t	left k_{t+1}	$\ln c_t$	$\beta^t \ln c_t$	TVC term
0	0.0500	0.9500	−2.9957	−2.9957	19.0000
1	0.0450	0.9050	−3.1011	−2.7910	18.1000
2	0.0405	0.8645	−3.2065	−2.5972	17.2900
3	0.0365	0.8280	−3.3118	−2.4143	16.5610
4	0.0328	0.7952	−3.4172	−2.2420	15.9049
5	0.0295	0.7657	−3.5225	−2.0800	15.3144
6	0.0266	0.7391	−3.6279	−1.9280	14.7830
7	0.0239	0.7152	−3.7333	−1.7856	14.3047
8	0.0215	0.6937	−3.8386	−1.6524	13.8742
9	0.0194	0.6743	−3.9440	−1.5280	13.4868
10	0.0174	0.6569	−4.0493	−1.4119	13.1381
∞	0	0.5000	—	—	10.0000

The Euler equation still holds: $c_{t+1}/c_t = 0.9 = \beta$. But the cake never disappears:

$$k_t \to 0.5, \qquad \beta^t u'(c_t)k_{t+1} = 20k_{t+1} \to 10 \neq 0.$$

The TVC fails, so this Euler-compliant plan cannot be optimal.

Exercise: dynamic programming with depreciation

If you do not eat the cake, it gets rotten a bit by a fraction of $\delta \in (0, 1)$. Find the optimal $\{c_t\}_{t\geq 0}$ to solve

$$\max_{\{c_t\}} \sum_{t=0}^{\infty} \beta^t \log c_t$$

subject to

$$k_{t+1} = (1-\delta)k_t - c_t, \qquad 0 < \delta < 1,$$

with $k_0 > 0$ and $0 \leq c_t \leq (1-\delta)k_t$.

1. Find the Euler equation by Lagrangian method
2. Find the Euler equation by Bellman method
3. Create spreadsheet detailing the optimal consumption and leftover cake, given $\beta = 0.9, \delta = 0.05, k_0 = 1$

Solution to the DP exercise

Optimal Control

The cake-eating problem in continuous time

Objective function

$$\max_{c(t)} \int_0^\infty e^{-\rho t} u(c(t))dt$$

subject to the dynamics

$$\dot{k}(t) = -c(t)$$

- Inititial condition $k(0) = k_0 > 0$
- Control: $c(t) \geq 0$

Some macro models are naturally written in continuous time. You will see a lot in labor economics and macroeconomics. Optimal control is the analog of dynamic programming in continuous time.

From discrete to continuous time

Start from:

$$\sum_{t=0}^{\infty} \beta^t u(c_t), \quad k_{t+1} = k_t - c_t$$

Introduce Δt such that $t = n\Delta t$. Since $0 < \beta < 1$, we can write the per-step discount factor as $\beta = e^{-\rho \Delta t}$.

Rescale so that stock = flow $\times \Delta t$

$$\sum_{t=0}^{\infty} \beta^t u(c_t) \ \rightarrow \ \sum_{n=0}^{\infty} e^{-\rho n \Delta t} u(c(n\Delta t))\Delta t$$

As $\Delta t \to 0$, take the limits (Riemann sum)

$$\lim_{\Delta t \to 0} \sum_{n=0}^{\infty} e^{-\rho n \Delta t} u(c(n\Delta t))\Delta t = \int_0^{\infty} e^{-\rho t} u(c(t))dt$$

Law of motion. From $k_{t+1} = k_t - c_t$ or $k(t+\Delta t) = k(t) - c(t)\Delta t$

$$\frac{k(t+\Delta t) - k(t)}{\Delta t} = -c(t)$$

Taking the limits: $\Delta t \to 0$ we obtain $\dot{k}(t) = -c(t)$

Halmintonian

To solve this problem, write the present-value Hamiltonian

$$\mathcal{H} = e^{-\rho t} u(c(t)) + \lambda(t) \Big[-c(t) \Big]$$

[∗] Note that $\lambda(t)$ is also called the co-state variable. The FOC:

$$\begin{aligned}
(c(t)) : \qquad & \frac{\partial \mathcal{H}}{\partial c} = e^{-\rho t} u'(c(t)) - \lambda(t) = 0, \\
(\dot{k}(t)) : \qquad & \dot{k}(t) = \frac{\partial \mathcal{H}}{\partial \lambda}, \\
(\dot{\lambda}(t)) : \qquad & \dot{\lambda}(t) = -\frac{\partial \mathcal{H}}{\partial k}
\end{aligned}$$

And the transversality condition

$$\lim_{t \to \infty} \lambda(t) k(t) = 0$$

In the cake-eating problem, we have:

$$e^{-\rho t}u'(c(t)) = \lambda(t)$$

State equation: $\dot{k}(t) = -c(t)$.
Co-state equation:

$$\dot{\lambda}(t) = 0$$

since $\mathcal{H}$ is independent of k. This implies $\lambda(t) = \lambda_0 \quad \forall t$.
Differentiate $\lambda(t)$ wrt time t, we obtain:

$$\dot{\lambda}(t) = -\rho e^{-\rho t}u'(c(t)) + e^{-\rho t}u''(c(t))\dot{c}(t)$$

But $\dot{\lambda}(t) = 0$ so

$$-\rho u'(c(t)) + u''(c(t))\dot{c}(t) = 0$$

Rearranging to obtain the Euler equation

$$u''(c)\dot{c} = \rho u'(c)$$

The transversality condition

$$\lim_{t\to\infty} \lambda(t)k(t) = \lim_{t\to\infty} e^{-\rho t}u'(c(t))k(t) = 0.$$

Solution

Assume that $u(c) = \log c$, then the Euler equation becomes

$$\frac{\dot{c}(t)}{c(t)} = -\rho$$

The LHS is actually

$$\frac{\dot{c}(t)}{c(t)} = \frac{d}{dt} \log c(t) = -\rho$$

Integrate both sides wrt t:

$$\int \frac{d}{dt} \ln c(t) dt = \int -\rho dt$$

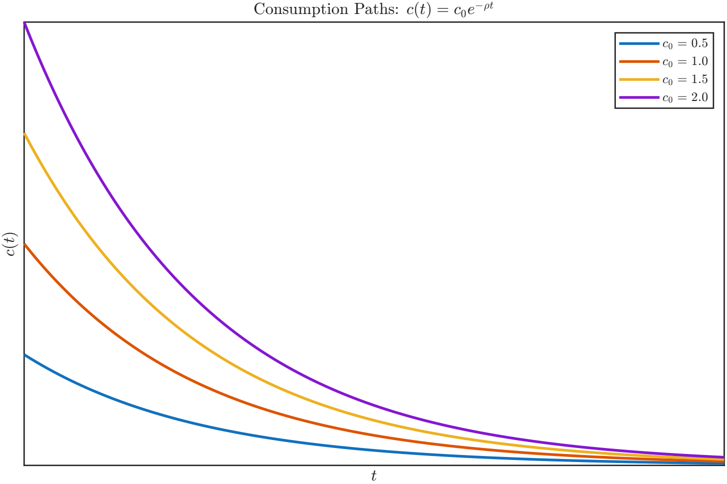
or

$$\ln c(t) = -\rho t + C$$

Exponentiate both sides

$$c(t) = e^{-\rho t + C} = e^{-\rho t} c_0. \qquad (c_0 \equiv e^C)$$

Time path



The HJB Equation

The Hamilton–Jacobi–Bellman (HJB) equation is the continuous-time analogue of Bellman ▸ Bellman to HJB

$$V(k) = \max_{c \geq 0} \{u(c) + \beta V(k')\}$$

In continuous time:

$$\rho V(k) = \max_{c \geq 0} \{u(c) + V'(k)\dot{k}\}$$

In the cake-eating problem ($\dot{k} = -c$):

$$\rho V(k) = \max_{c \geq 0} \{u(c) - cV'(k)\}$$

Interpretation:

- LHS: required return on the value function
- RHS: flow utility + change in value of the state

From HJB to Dynamics of $V'(k)$

HJB:

$$\rho V(k) = \max_c \{u(c) - cV'(k)\}$$

First-order condition:

$$u'(c) = V'(k)$$

Differentiate the HJB with respect to k:

$$\rho V'(k) = \frac{d}{dk}[u(c) - cV'(k)]$$

Using the chain rule:

$$\rho V'(k) = u'(c)\frac{\partial c}{\partial k} - \left[\frac{\partial c}{\partial k}V'(k) + cV''(k)\right]$$

Using $u'(c) = V'(k)$, the first two terms cancel:

$$\rho V'(k) = -cV''(k)$$

Euler Equation from HJB

From the previous slide:

$$\rho V'(k) = -cV''(k)$$

Differentiate wrt time

$$\dot{V}'(k) = V''(k)\dot{k}$$

Since $\dot{k} = -c$:

$$\dot{V}'(k) = -cV''(k)$$

Substitute:

$$\dot{V}'(k) = \rho V'(k)$$

Using the FOC $V'(k) = u'(c)$:

$$\dot{u}'(c) = \rho u'(c)$$

For $u(c) = \log c$, $u'(c) = \frac{1}{c}$:

$$-\frac{1}{c^2}\dot{c} = \rho\frac{1}{c} \Rightarrow \quad \frac{\dot{c}}{c} = -\rho$$

Exercise: optimal control with regeneration

A renewable resource stock evolves according to

$$\dot{k}(t) = ak(t) - c(t), \qquad a > 0,$$

and the planner solves

$$\max_{c(t)} \int_0^\infty e^{-\rho t} \log c(t)\, dt$$

subject to $k(0) = k_0 > 0$ and $c(t) \geq 0$.

1. Write the present-value Hamiltonian.
2. Derive the FOC, state equation, and co-state equation.
3. Show that the optimal consumption growth rate satisfies an Euler equation of the form $\dot{c}/c =$ constant, and identify that constant.
4. Use the TVC to discuss the restriction on parameters needed for an economically meaningful solution.

▸ Solution to the OC exercise

Differential Equations and Phase Diagrams

Autonomous ODEs and steady states

An **autonomous** ordinary differential equation is

$$\dot{x}(t) = f(x(t)),$$

where the right-hand side does not depend explicitly on time.

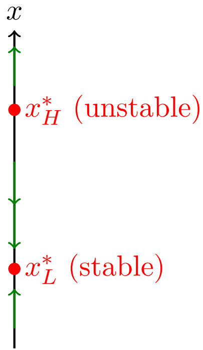
- A **steady state** (equilibrium) is a point x^* with $f(x^*) = 0$.
- x^* is locally stable if solutions starting near x^* converge to it; unstable if they move away.
- In macroeconomics we care about the *qualitative* behavior—convergence, divergence, cycles—not closed-form solutions.

Examples: $\dot{x} = \lambda x$ (exponential growth or decay) and $\dot{x} = rx(1 - x/K)$ (logistic growth).

One variable: the phase line

For $\dot{x} = f(x)$, draw the phase line:

1. Find the steady states: solve $f(x) = 0$.
2. Determine the sign of $\dot{x}$ on each interval between steady states.
3. Draw arrows: $\dot{x} > 0$ points up, $\dot{x} < 0$ points down.



Near a steady state, linearize: $f(x) \approx f'(x^*)(x - x^*)$. Hence $f'(x^*) < 0$ $\Rightarrow$ stable, and $f'(x^*) > 0 \Rightarrow$ unstable.

Two variables: linear systems

Consider

$$\begin{cases} \dot{x} = ax + by, \\ \dot{y} = cx + dy, \end{cases} \qquad \dot{\mathbf{x}} = A\mathbf{x}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The steady state is $(0,0)$; its stability is determined by the eigenvalues λ_1, λ_2 of A:

- Sink (stable): $\lambda_1, \lambda_2 < 0$ (equivalently $\Delta > 0$, $\tau < 0$).
- Source (unstable): $\lambda_1, \lambda_2 > 0$ (equivalently $\Delta > 0$, $\tau > 0$).
- Saddle: eigenvalues of opposite sign (equivalently $\Delta < 0$).
- Complex $\lambda = \alpha \pm i\beta$: focus or centre, stable if $\alpha < 0$.

Here $\tau = \mathrm{tr}(A)$ and $\Delta = \det(A)$. **In macro, the saddle case is the relevant one.**

How to draw a phase diagram

For a two-dimensional system $\dot{k} = G(k, c)$, $\dot{c} = H(k, c)$:

1. Solve the **nullclines** $\dot{k} = 0$ and $\dot{c} = 0$.
2. Find the steady state at their intersection.
3. Determine the sign of $\dot{k}$ and $\dot{c}$ in each region.
4. Add horizontal and vertical arrows, then combine them into trajectories.
5. Identify the **stable arm** and the transition path from the initial state.

Useful trick: write, for example, $\dot{k} = b\,(c - c_k(k))$, where $c = c_k(k)$ is the $\dot{k} = 0$ locus. Then only the sign of the coefficient b matters for the direction of motion across the locus.

Worked example: a linear saddle

Consider

$$\dot{x} = ax + by + \kappa_1, \qquad \dot{y} = cx + dy + \kappa_2,$$

with $a < 0$, $b < 0$, $c < 0$, $d > 0$, $\kappa_1, \kappa_2 > 0$.

$$|A| = ad - bc < 0 \quad \Longrightarrow \quad \text{the steady state is a saddle.}$$

The nullclines are

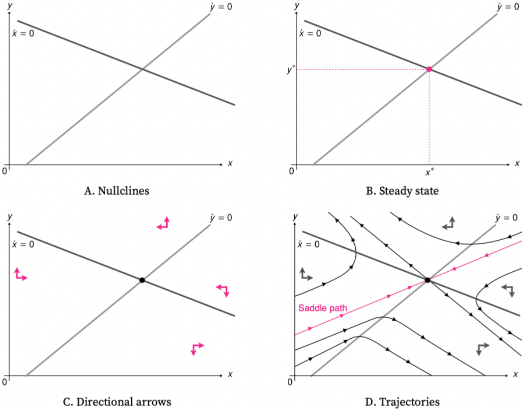
$$\dot{x} = 0: \quad y = -\frac{a}{b}x - \frac{\kappa_1}{b} \quad \text{(negative slope)},$$

$$\dot{y} = 0: \quad y = -\frac{c}{d}x - \frac{\kappa_2}{d} \quad \text{(positive slope)}.$$

The steady state solves

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} -\kappa_1 \\ -\kappa_2 \end{pmatrix}.$$

Worked example: the phase plane


y
ẏ = 0
ẋ = 0
0
x
A. Nullclines
y
ẏ = 0
ẋ = 0
y*
0
x*
x
B. Steady state
y
ẏ = 0
ẋ = 0
0
x
C. Directional arrows
y
ẏ = 0
ẋ = 0
Saddle path
0
x
D. Trajectories

- The nullclines divide the plane into four regions.
- Trajectories approach the saddle along the stable arm and diverge along the unstable arm.
- ▸ How to derive the stable arm

Nonlinear systems: linearization

Consider $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ with steady state $(\bar{x}, \bar{y})$. Linearize:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} \approx J(\bar{x}, \bar{y}) \begin{pmatrix} x - \bar{x} \\ y - \bar{y} \end{pmatrix}, \qquad J = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(\bar{x}, \bar{y})}.$$

Let $\tau = \mathrm{tr}(J)$ and $\Delta = \det(J)$:

- $\Delta > 0$, $\tau < 0$: stable node or focus.
- $\Delta > 0$, $\tau > 0$: unstable node or focus.
- $\Delta < 0$: saddle (one stable and one unstable direction).
- Complex roots $\alpha \pm i\beta$: stable focus if $\alpha < 0$, unstable if $\alpha > 0$, centre if $\alpha = 0$.

Application: the Ramsey phase diagram

The linearized Ramsey system is

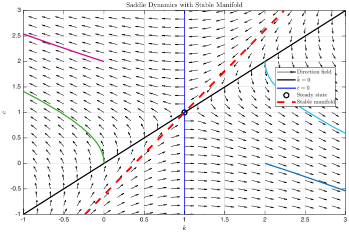
$$\dot{k} = a(k - k^*) - (c - c^*), \qquad \dot{c} = -b(k - k^*), \qquad a, b > 0.$$

The Jacobian is

$$J = \begin{pmatrix} a & -1 \\ -b & 0 \end{pmatrix}, \qquad \det(J) = -b < 0.$$

Hence the steady state is a saddle: there is one stable and one unstable eigenvalue.

- k is predetermined; c is a jump variable.
- Given k_0, there is a unique c_0 that places the economy on the stable arm.
- This is exactly the saddle-path logic used in the Ramsey model.



What you need to remember

- An autonomous ODE is summarized by its steady states and the sign of $\dot{x}$ between them.
- In two dimensions, compute the Jacobian and use (τ, Δ): $\Delta < 0$ means a saddle.
- To draw a phase diagram: nullclines → signs → arrows → stable arm.
- Macro models typically have one stable and one unstable direction; the stable arm pins down the jump variable.
- Discrete-time models (RBC) use log-linearization and the Blanchard–Kahn condition instead of continuous-time phase diagrams.

APPENDIX

Guess and Verify

The Bellman becomes

$$V(k) = A + B \log k = \max_c \log c + \beta[A + B \log(k - c)]$$

The FOC

$$\frac{1}{c} = \frac{\beta B}{k - c}$$

implying

$$\begin{aligned} c^* &= k/(1 + \beta B), \\ k' &= k - c^* = k - \frac{k}{1 + \beta B} = \frac{\beta B}{1 + \beta B} k \end{aligned}$$

This means:

$$\begin{aligned} \log c^* &= \log k - \log(1 + \beta B), \\ \log(k - c^*) &= \log k + \log\left(\frac{\beta B}{1 + \beta B}\right) \end{aligned}$$

Substituting all to the Bellman equation

$$V(k) = \log k - \log(1+\beta B) + \beta A + \beta B[\log k + \log(\beta B/(1+\beta B))]$$

Matching coefficients

$$(\log k) : B = 1 + \beta B \implies B = 1/(1-\beta)$$

The constant part

$$A(1-\beta) = -\log(1+\beta B) + \beta B \log(\beta B/(1+\beta B))$$

Plugging in $B = 1/(1-\beta)$ and solve for A yields

$$A = \frac{\log(1-\beta)}{1-\beta} + \frac{\beta \log \beta}{(1-\beta)^2}$$

Solution manual: DP exercise

Problem:

$$V(k) = \max_{0 \le c \le (1-\delta)k} \left\{ \log c + \beta V\Big((1-\delta)k - c\Big) \right\}.$$

FOC and envelope:

$$\frac{1}{c_t} = \beta V'(k_{t+1}), \qquad V'(k_t) = \frac{1-\delta}{c_t}.$$

Shifting the envelope one period gives $V'(k_{t+1}) = (1-\delta)/c_{t+1}$, hence

$$\frac{1}{c_t} = \beta \frac{1-\delta}{c_{t+1}} \quad \Rightarrow \quad c_{t+1} = \beta(1-\delta)c_t.$$

Guess $c_t = \alpha k_t$. Then

$$k_{t+1} = (1-\delta-\alpha)k_t, \qquad \alpha(1-\delta-\alpha) = \beta(1-\delta)\alpha.$$

For $\alpha > 0$, we get $\alpha = (1-\beta)(1-\delta)$ and thus the policy:

$$c_t = (1-\beta)(1-\delta)k_t, \qquad k_{t+1} = \beta(1-\delta)k_t.$$

Solution manual: TVC and optimal control exercise

For the DP exercise, the policy is

$$c_t = (1-\beta)(1-\delta)k_t, \qquad k_t = [\beta(1-\delta)]^t k_0.$$

Then

$$\beta^t u'(c_t)k_{t+1} = \beta^t \frac{1}{c_t} k_{t+1} = \beta^t \frac{\beta(1-\delta)k_t}{(1-\beta)(1-\delta)k_t} = \frac{\beta^{t+1}}{1-\beta} \to 0,$$

so the TVC holds because $\beta < 1$.

Solution for the Optimal Control

For the optimal control exercise:

$$\mathcal{H} = e^{-\rho t} \log c(t) + \lambda(t)[ak(t) - c(t)].$$

Necessary conditions are

$$e^{-\rho t} \frac{1}{c(t)} = \lambda(t), \qquad \dot{k} = ak - c, \qquad \dot{\lambda} = -a\lambda.$$

Differentiating $\lambda = e^{-\rho t}/c$ wrt time and equating to $-a\lambda$ gives

$$\frac{\dot{c}(t)}{c(t)} = a - \rho.$$

Thus consumption grows at rate $a - \rho$ (positive iff $a > \rho$). With log utility the budget gives $c(t) = \rho k(t)$, so k also grows at $a - \rho$ and $\lambda(t)k(t) = e^{-\rho t}k(t)/c(t) = e^{-\rho t}/\rho \to 0$: the TVC holds for any $\rho > 0$, and lifetime utility is finite.

From Discrete Bellman to Continuous Time (1–3)

Start from the discrete-time Bellman equation:

$$V(k_t) = \max_{c_t} \{u(c_t) + \beta V(k_{t+1})\}$$

subject to:

$$k_{t+1} = k_t - c_t$$

Introduce a small time step Δt:

- Flow utility: $u(c_t)\Delta t$
- Discounting: $\beta = e^{-\rho \Delta t}$
- Law of motion: $k_{t+\Delta t} = k_t - c_t \Delta t$

Rewrite Bellman equation:

$$V(k) = \max_{c} \left\{ u(c)\Delta t + e^{-\rho \Delta t} V(k - c\Delta t) \right\}$$

Taylor Expansion and Rearrangement (4–7)

Use first-order approximations:

$$V(k - c\Delta t) \approx V(k) - c\Delta t\, V'(k)$$
$$e^{-\rho\Delta t} \approx 1 - \rho\Delta t$$

Substitute into Bellman:

$$V(k) = \max_c \left\{ u(c)\Delta t + (1 - \rho\Delta t)\Big[V(k) - c\Delta t V'(k)\Big] \right\}$$

Expand:

$$V(k) = \max_c \Big[u(c)\Delta t + V(k) - c\Delta t V'(k) - \rho\Delta t V(k) \Big]$$

Cancel $V(k)$:

$$0 = \max_c \left[u(c)\Delta t - c\Delta t V'(k) - \rho\Delta t V(k) \right]$$

Continuous-Time Limit: HJB Equation (8–9)

Divide both sides by Δt:

$$0 = \max_c \left[u(c) - cV'(k) - \rho V(k)\right]$$

Rearrange:

$$\rho V(k) = \max_c \left\{u(c) - cV'(k)\right\}$$

Using $\dot{k} = -c$, we obtain the HJB:

$$\rho V(k) = \max_c \left\{u(c) + V'(k)\dot{k}\right\}$$

Key insight:

- Bellman: $V(k) = u(c) + \beta V(k')$
- HJB: $\rho V = u + V'(k)\dot{k}$

The Stable Arms

One can write the system as

$$\dot{z} = Az + \kappa, \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Deviations near the steady state is

$$\Delta z = z - z^*$$

so

$$\dot{\Delta} z = A \Delta z$$

Characteristic equation

$$\det(A - \lambda I) = 0$$

yields the eigenvalues as solutions to:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Assume that there are two solutions $\lambda_1 > 0 > \lambda_2$

For each eigenvalue λ, the eigenvector $\mathbf{v}$ satisfies $(A - \lambda I)\mathbf{v} = 0$. Since $b \neq 0$, the convenient choice is

$$\mathbf{v}(\lambda) = \begin{pmatrix} b \\ \lambda - a \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{\lambda - a}{b} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{a-\lambda}{b} \end{pmatrix}$$

Then we can write Δz rigorously

$$\Delta z(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2, \qquad \lambda_1 > 0, \lambda_2 < 0$$

Since $\lambda_1 > 0$, convergence to the steady state requires $c_1 = 0$. Set $c_1 = 0$, at $t = 0$, we have $\Delta z(0) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ (here, c_1, c_2 are the coordinates of the initial deviation). Set explosive path $c_1 = 0$, we want to derive the value of y that "jumps" to the stable arm whenever there is deviation:

$$\Delta z = c_2\mathbf{v}_2,$$

meaning

$$\Delta z = \begin{pmatrix} x - x^* \\ y - y^* \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{a-\lambda_2}{b} \end{pmatrix} \text{ or } y - y^* = -\frac{a - \lambda_2}{b}(x - x^*)$$

which is the policy function on the stable manifold

Further Readings

- Stokey, Nancy L., Robert E. Lucas, and Edward C. Prescott. Recursive Methods in Economic Dynamics. (1989). Harvard University Press.
- Essential mathematics for economic analysis (Sydsaeter & Hammond et al., 6e 2021)
- Further mathematics for economic analysis (Sydsaeter & Hammond et al., 2e 2008)
- Mathematics for economists (Simon & Blume, 1994)